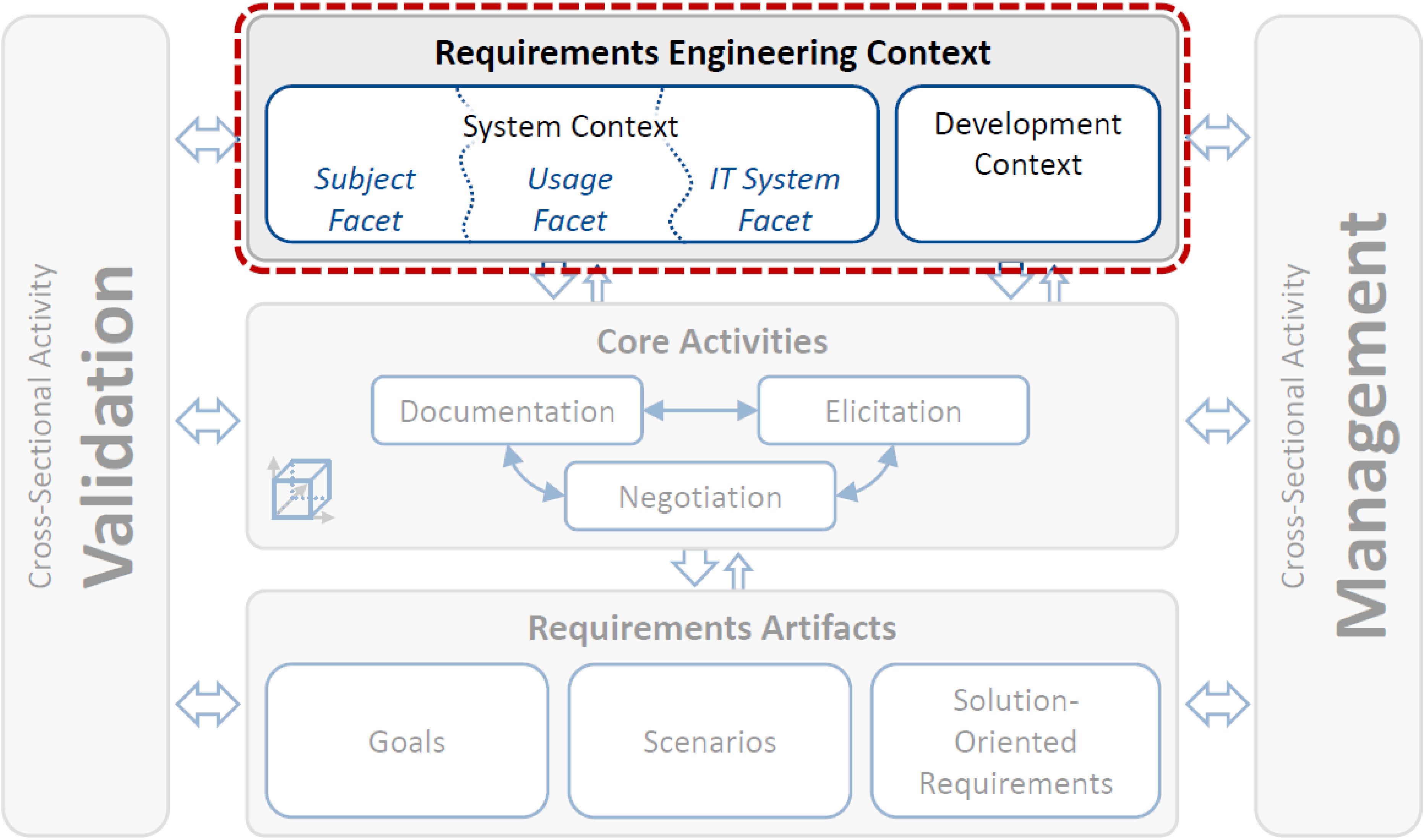
CKS121/CSE 242 Software Requirements

Context (Part 1)

Requirements Engineering Framework



Learning outcomes

The students

- understand the importance of considering the context of a system during requirements engineering and system development.
- learn about the system context, its objects and relationships.
- get a heuristic for structuring the system context.
- understand the system context changes over time.

Structure of this Lecture

- **Context of a System**
- System Context and Context Objects
- Change of System Context
- Consideration of System Context Objects
- Three Facets of System Context
- Properties of System Context Objects
- Documentation of Context Information

Context of a (Software-intensive) System

- A (software-intensive) system is always **embedded in a particular context**.
- The **context** heavily **influences the requirements** the system has to fulfill.
- **A requirement is always defined for a particular context**.

Revisit!

Accounting System (1)

Example: Develop an accounting system for the manufacturing industry in Germany.

In this context we have to consider

- relevant **German laws** and
- business needs of **manufacturing industry**.

Accounting System (2)

Example: Develop an accounting system for financial companies in the USA.

In this context we have to consider

- relevant **US American laws** and
- business needs of **financial companies**.

Accounting System (3)

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Example: Develop an accounting system for a shipping company in China.

In this context we have to consider

- relevant **Chinese laws** and
- business needs of **shipping companies**.

Wind Turbine (1)

E

Example: Develop a power supply, based on renewable energies.

Possible solution: Wind turbine

Context: Family home in a countryside

To produce power in a windy place for a family home, a wind turbine can be a **good solution**.

Wind Turbine (2)

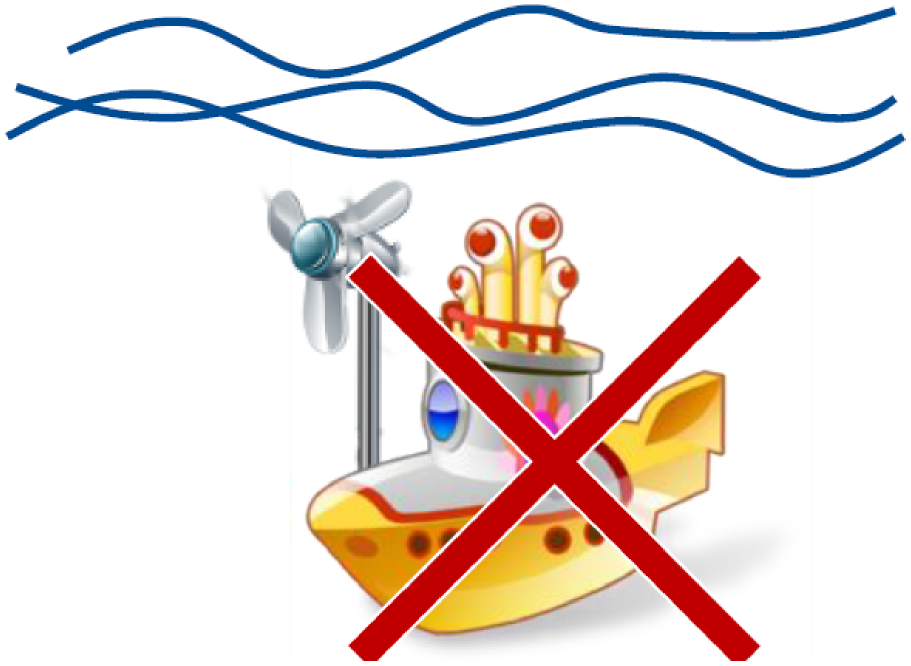
E

Example: Develop a power supply, based on renewable energies.

Possible solution: Wind turbine

Context: Submarine under water

To produce power for a submarine under water, a wind turbine is **an inappropriate solution**.



Wind Turbine (3)

E

Example: Develop a power supply, based on renewable energies.

Possible solution: Wind turbine

Context: Satellite in space

To produce power for a satellite in space,
a wind turbine is **an inappropriate solution**.

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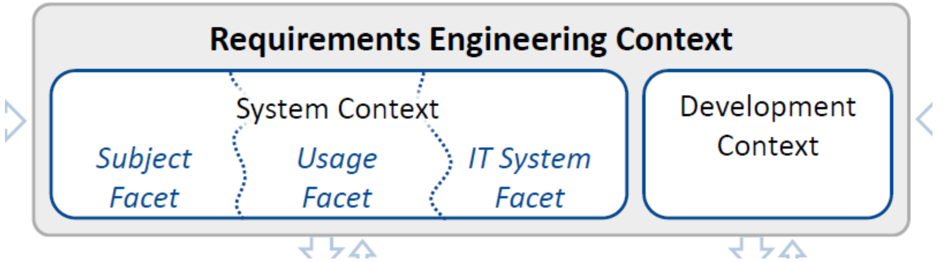
Context Objects

D Context objects are **material or immaterial objects** belonging to the context.

Typical examples of context object types are:

Material Objects (Can be touched!)	**Immaterial Objects** (Cannot be touched!)
• People • Hardware • Documents (manuals, standards, laws, …) • Buildings • Cars • …	• Organizations • Business processes • Software components • Data • Communication services • …



System Context and Context Objects

System Context

D The system context is the part of the context in which the system to be developed **is operating / embedded**.

D Material or immaterial objects belonging to the system context are called **system context objects**.

System context objects are relevant for the system to be developed and thus **have to be considered during requirements engineering**.

System Context Objects

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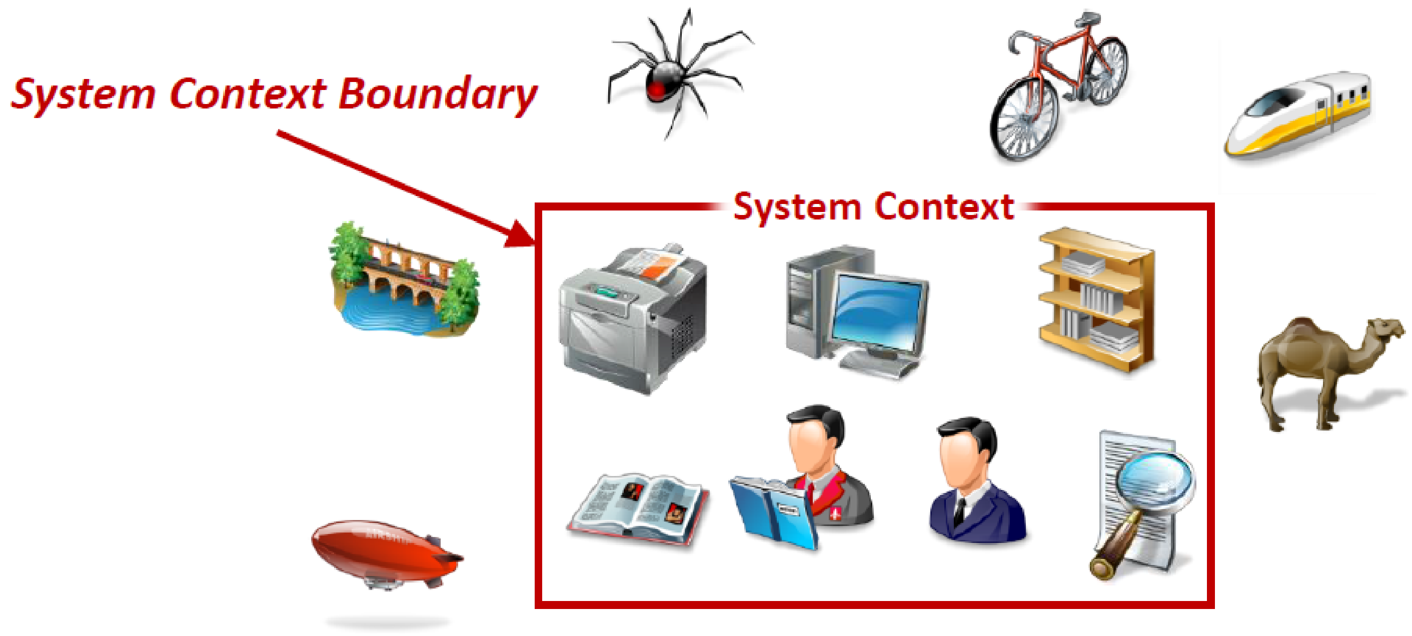
Typical examples for system context objects of a university library system are:

Material Objects (Can be touched!)	Immaterial Objects (Cannot be touched!)
• Books • Book shelf • Marc Genaro (a student) • Jennifer Adrian (an employee) • Workstation • Printer • …	• Library database • Meta-search engine accessing several library systems • User authorization service • …

System Context Boundary

D The **system context boundary** defines which material and immaterial objects belong to the system context. It thereby **separates system context objects from context objects**.



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Change of System Context Scope (1)

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"Establish a fast and safe transportation!"



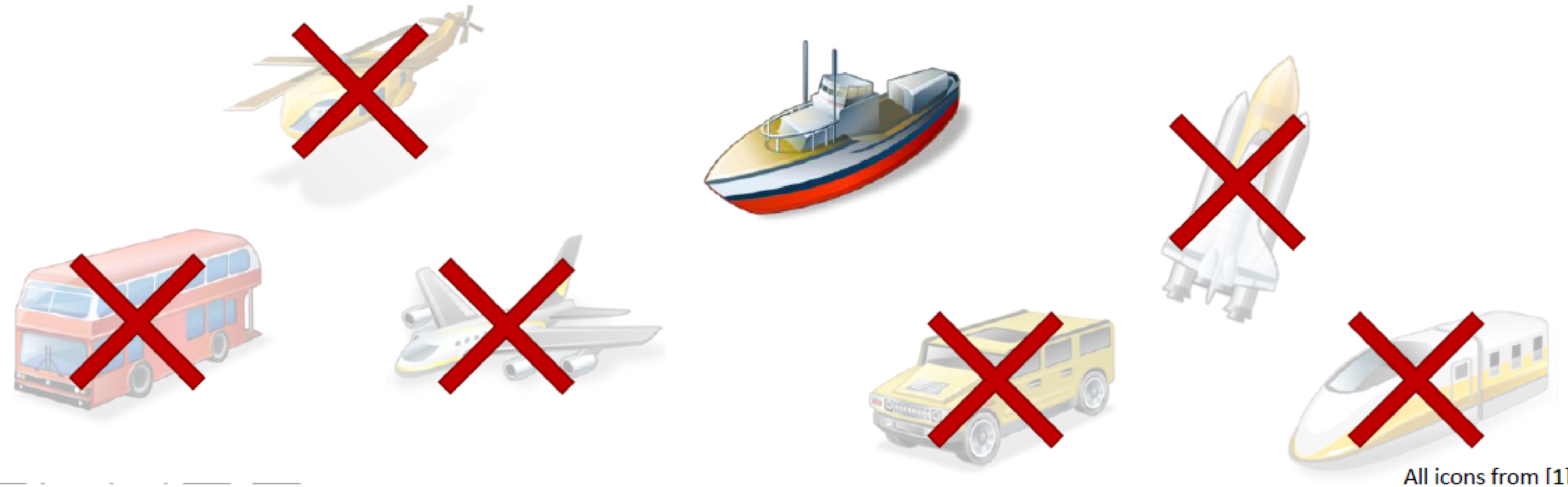

All icons from [1]

Change of System Context Scope (2)

E

During elicitation the requirements engineer elicits more information

- "The transportation should be between the mainland and an island."
- "The island has about 35 inhabitants."
- "The island is located 5 km from the mainland."



All icons from [1]

Change of System Context Scope (3)

- One stakeholder suggests to use a **marine diesel** engine.
- Another stakeholder suggests **to avoid** as far as possible **any kind of pollution**, since transportation crosses a natural reserves.
- The decision to use an **electric engine** resolves this conflict.

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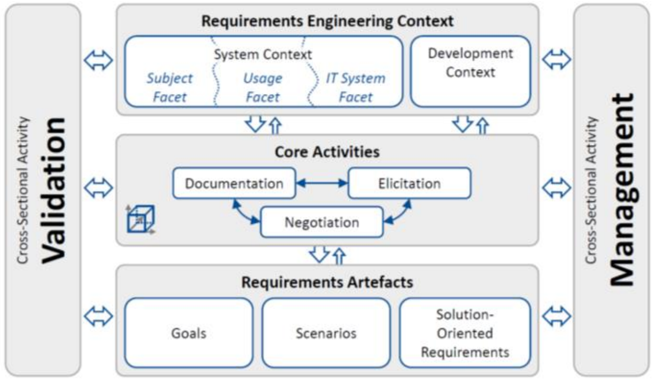
Involvement in Requirements Engineering Activities

- **Each context object can potentially** be **involved in any** requirements engineering **activities**.
  - Elicitation, documentation and negotiation.
  - Validation and management.
- If a context object is not involved in a requirements engineering activity, it
  - might still be relevant for this requirements engineering activity and
  - might be relevant for all other requirements engineering activities.



Consideration of System Context Objects

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- Assume, there are **10 stakeholders** in the system context (e.g., office worker) which potentially have **relevant information for defining the requirements** for the system to be developed.
- Due to time restrictions **only 6** out of 10 potential relevant stakeholders were **interviewed**.
- The **remaining 4 stakeholders are still potentially relevant** for later interviews and/or other requirements engineering activities.

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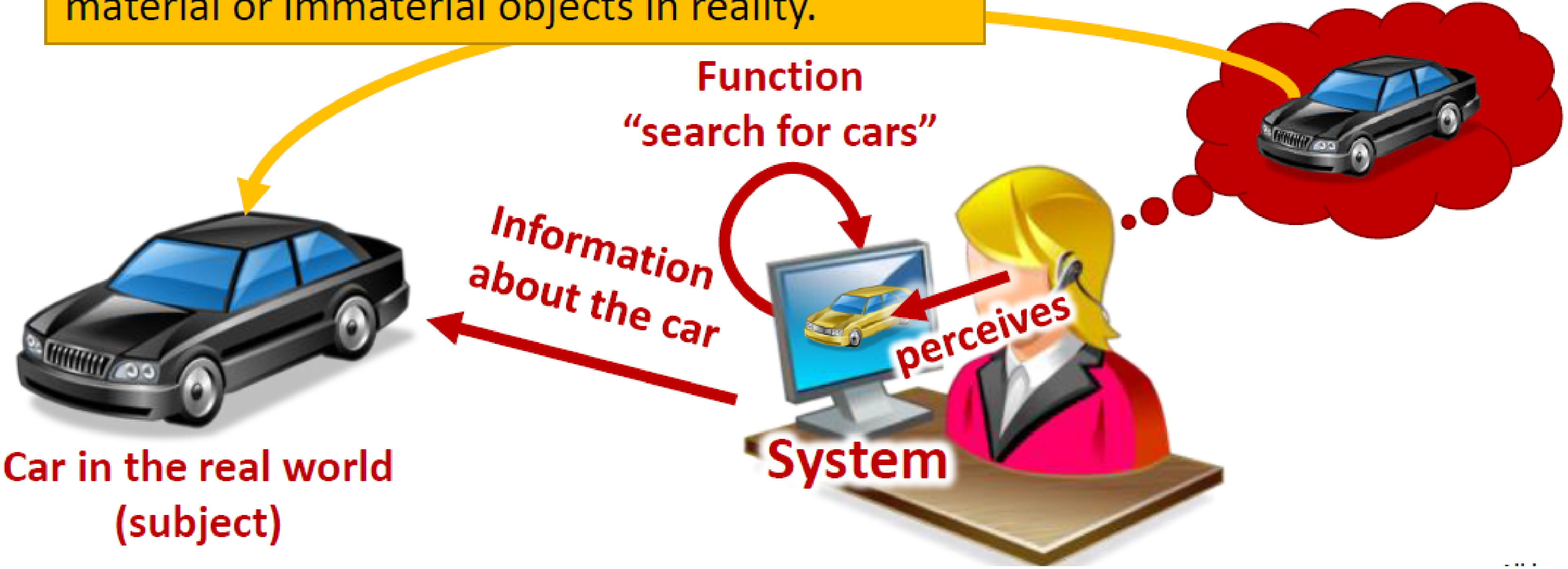
Three Facets of System Context

Generic Principles of any Software-System

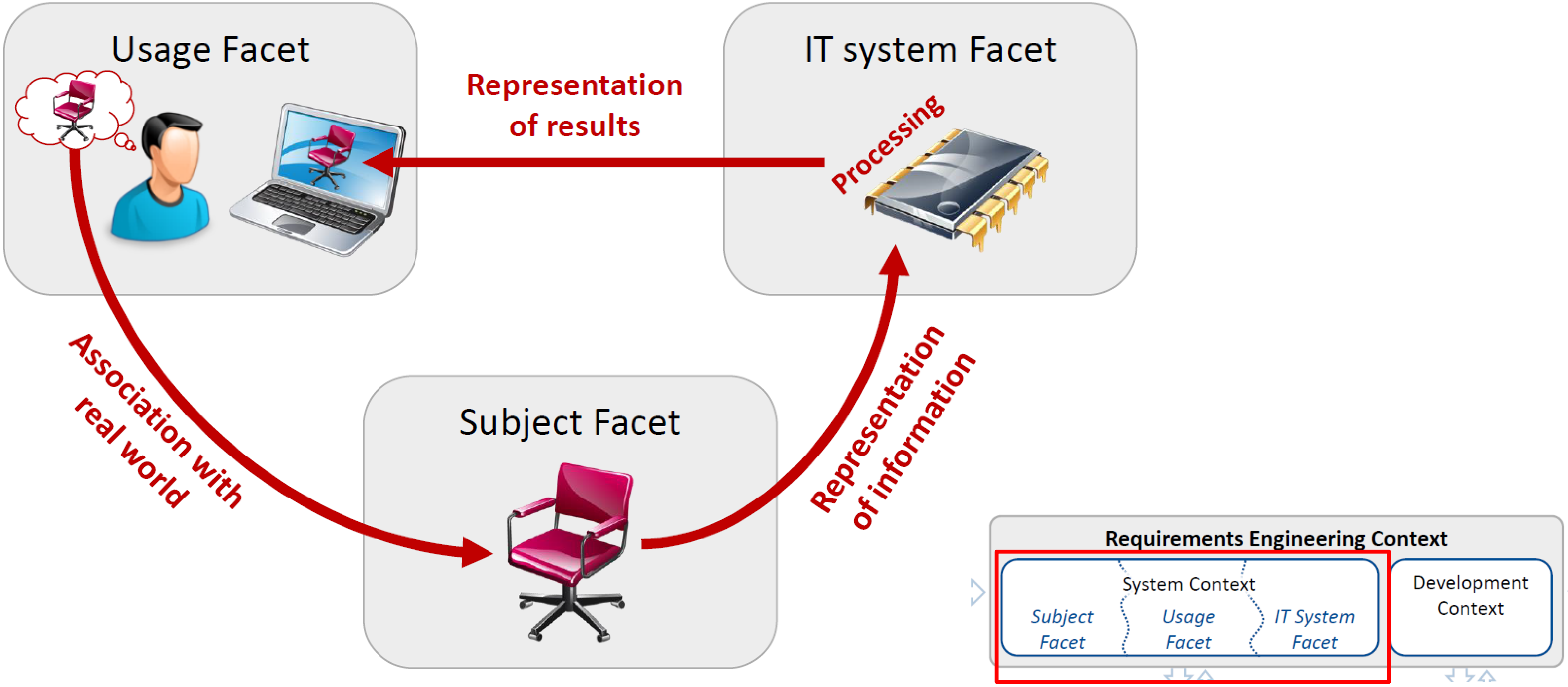
Information systems as well as embedded systems

- represent information about the real world,
- process this information and provide some functionality, e.g., search for car types,
- provide an output to the context, e.g., displays the retrieved cars.

System is only successful if the user can map the information displayed to the corresponding material or immaterial objects in reality.

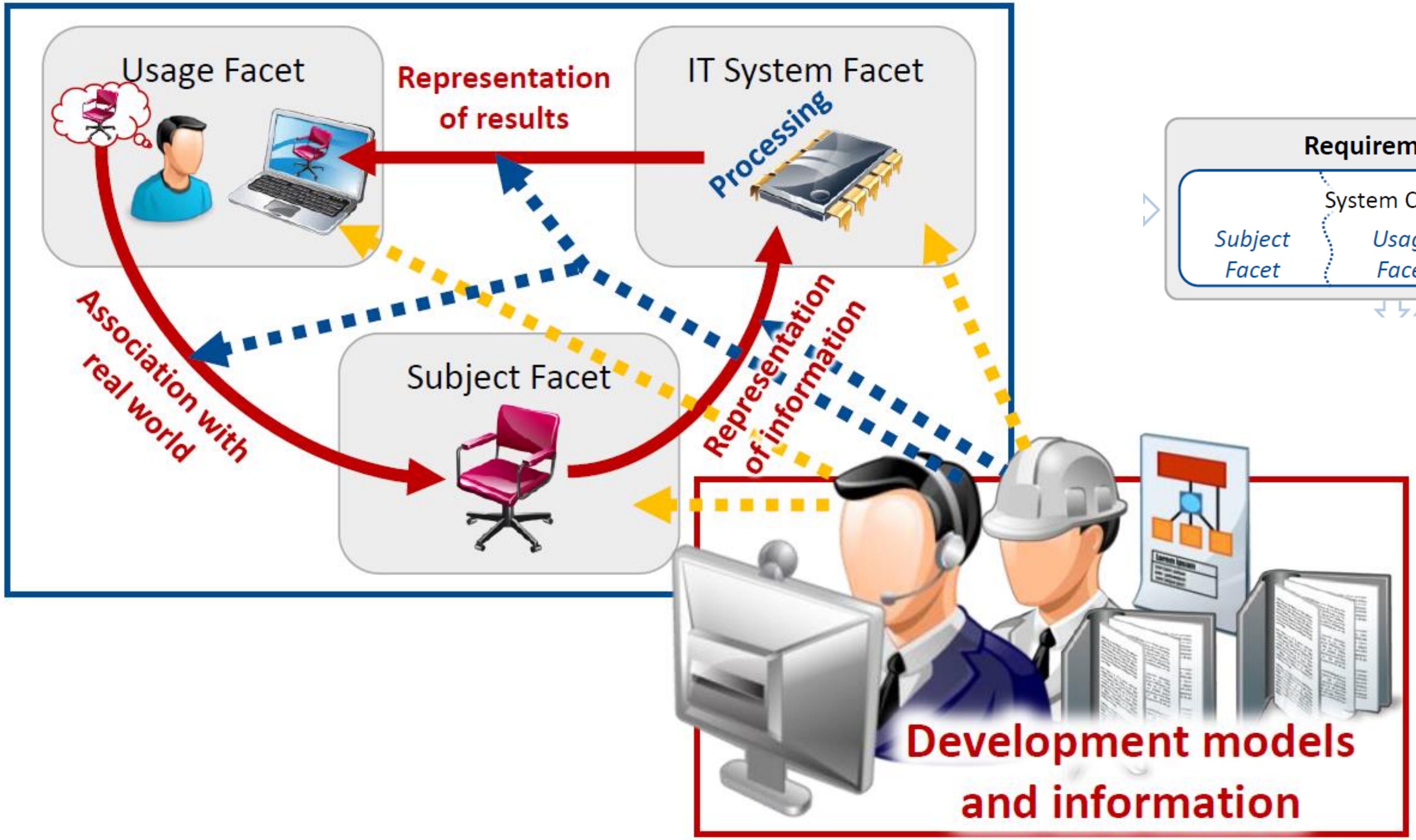


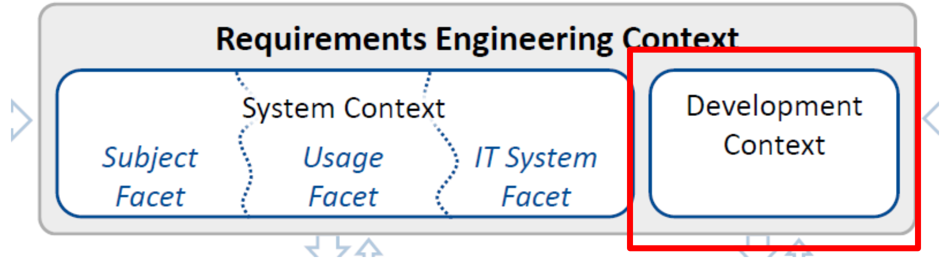
Three Facets of System Context



Three Facets of System Context

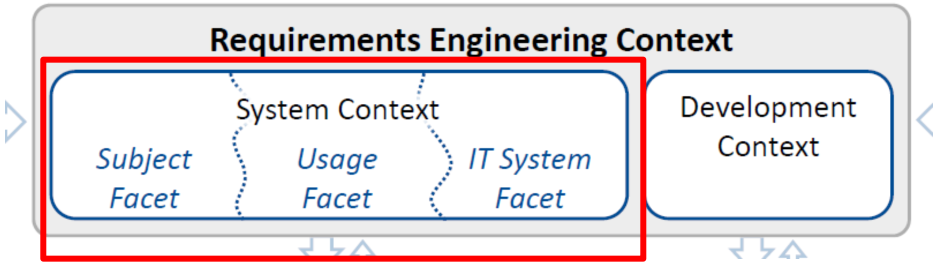
Development Context



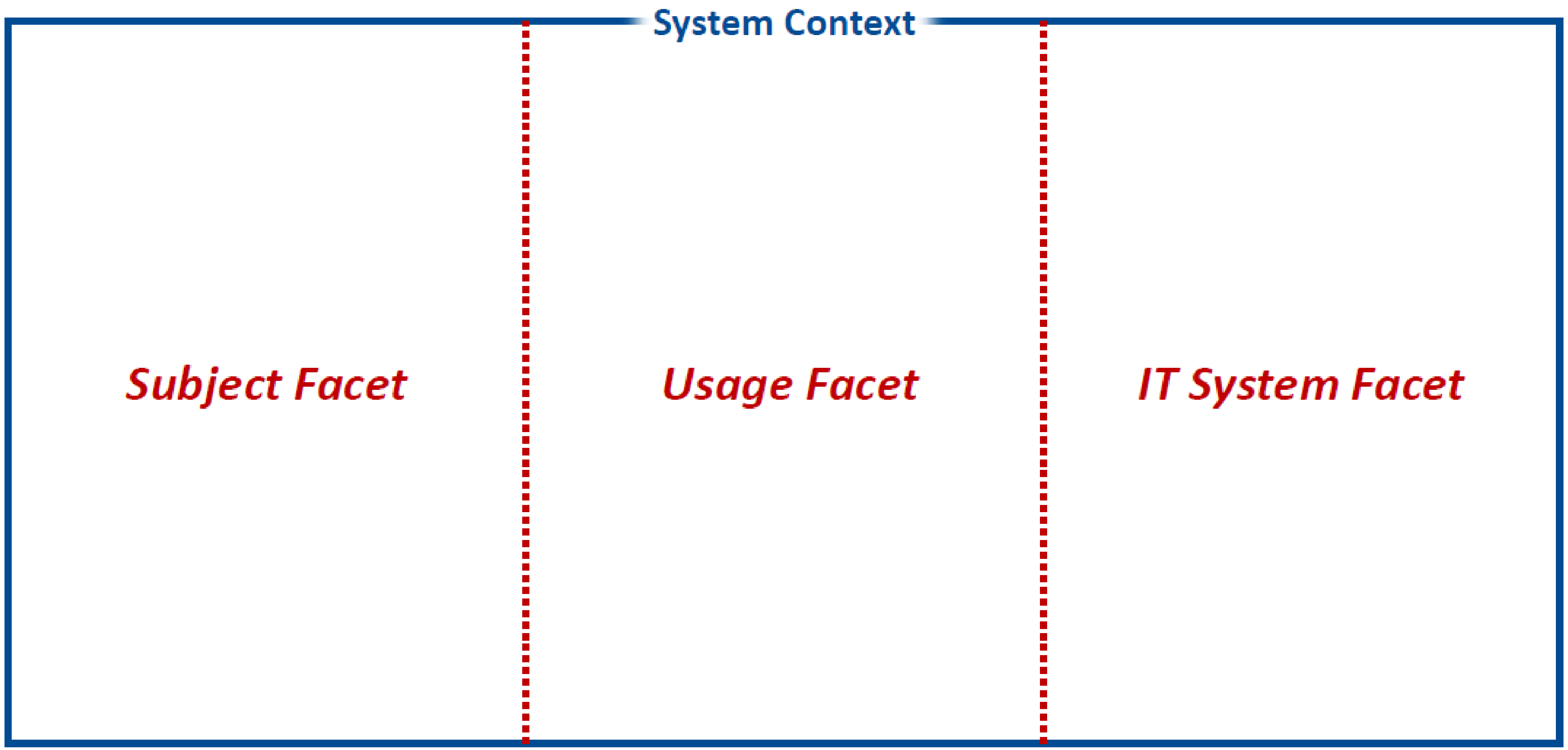


Three Facets of System Context

Structuring the System Context



We differentiate between **three system context facets**.



Subject Facet (1)

D The **subject facet** comprises system context objects about which **information is represented in the system** or which influence or constrain the representation of information represented in the system.

This includes system context **objects and events**, their **properties and relationships** about which information is represented in the system and the **quality of the representation** such as accuracy and actuality.

Subject Facet (2)

E

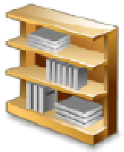
Typical examples for system context objects in the subject facet are:

- Lawyers and data privacy officers
- Reference models of the subject domain
- Textbooks and laws
- People about which data is stored (e.g., customers, suppliers)
- Consumer goods
- Processes (e.g., production or business processes supported or automated by the system)

Subject Facet (3)

<u>Example</u>: Development of a university library system.

Typical examples for system context objects in the subject facet are:

- Student 
- Book 
- Magazine
- E-book
- Bookshelf 
- …

Usage Facet (1)

D The **usage facet** comprises all **system context objects** (people and/or systems) which directly or indirectly **interact with the system** or which **influence or benefit from the usage** of the system.

This includes system context **objects** contributing to the definition of the **desired usage** and the **usage quality attributes** such as expected usage load or average response times.

Usage Facet (2)

E

Typical examples for system context objects in the usage facet are:

- Usage goals and tasks
- Workflows or business processes
- User groups
- Interaction models (with other systems)
- Interfaces
- Usage laws and standards

Usage Facet (3)

E

<u>Example</u>: Development of a university library system.

Typical examples for system context objects in the usage facet are:

- Student user group
- Employee user group
- Meta-search engine
  accessing several library systems

- Check-out process and return process for library items
- ...

IT System Facet (1)

D The <u>IT system facet</u> comprises all <u>system context objects</u> of the <u>technical and operational environment</u> in which the system is going to be deployed in or which <u>influence or constraint</u> the deployment of the system and/or the <u>use of technology</u> by the system such as sensors, actuators or other systems.

This includes system context <u>objects</u> which contribute to the definition of the <u>operational environment and/or the technology</u> used or which influence or constraint relevant <u>quality attributes</u> (such as average availability time, security).

IT System Facet (2)

E

Typical examples for system context objects in the IT system facet are:

- Hard- and software platforms
- Communication networks
- Peripheral devices
- Hardware components
- Other software-based systems
- IT policies and IT strategies
- Infrastructure documents

IT System Facet (2)

E

<u>Example</u>: Development of a university library system.

Typical examples for system context objects in the IT system facet are:

- Printer

- Server
- Library workstations

- University cloud-infrastructure
- ...

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Consideration of Properties

Not only the system context objects within each of the three system context facets, but also their

- properties,
- relationships with other objects in the same facet and
- relationships with objects in the other facets

have to be considered during requirements engineering.

Subject Facet

E

Examples for **properties of system context objects in the subject facet** for an university library system are:

System Context Object	Properties
Student	• ID • Matriculation status • Date of birth
Book	• Title • Authors • ISBN • ISSN
Data privacy law	• Visibility of data for librarian

Usage Facet

E

Examples for **properties of system context objects in the usage facet** for an university library system are:

System Context Object	Properties
Students (user group)	• Maximum number of users • Average usage load
Employees (as user group)	• Required access control and usage rights
Meta-search engine	• Required information for a particular usage

IT System Facet

E

Examples for **properties of system context objects in the IT system facet** for an university library system are:

System Context Object	Properties
University cloud-infrastructure	• Maximum throughput of communication network
Server	• Maximum persistent storage available • Maximum downtime in 24h

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Requirements and Context Information

- A requirement is **always defined for a particular context**.
- Consequently to understand requirements, context information is required.
- A **change in the context** typically **requires adaptation of requirements**.
- **Documenting context information** (context objects, their properties and relationships) is a **prerequisite for supporting the analysis** of the **impact of a context change on requirements**!

Problems

- Context information is **often not documented at all** and therefore only implicitly known.

  Different stakeholders might thus have different assumptions about the context.

- **If documented**, the documentation of context information and requirements is **often intermingled**. I.e. context information is spread across requirements. This leads to:
  - Redundancies of context information.
  - Not concise definition of context information.
  - Contradictions between context information.

Guideline for Documenting Context Information

- Define <u>project-specific guidelines</u> for documenting context information.
- The guidelines should include, for example:
  - Types of context objects which should be considered and documented.
  - Representation formats and the structure to be used.
  - Relationship types to interrelate context information and requirements.
  - Roles and responsibilities for documenting the context information.

Common Approaches

- **Scenarios**: Define system context interaction and thereby interrelate requirements with system context objects.
- **(Business) process models**: Define the business context in which system functions are used.
- **Domain models**: Define properties for a set of systems, e.g., for systems in a particular application domain.
- **Standards**: Include context information such as personal data protection, federal ordinance on barrier-free information technology.
- Dedicated section in textual requirements specification documents.
- …

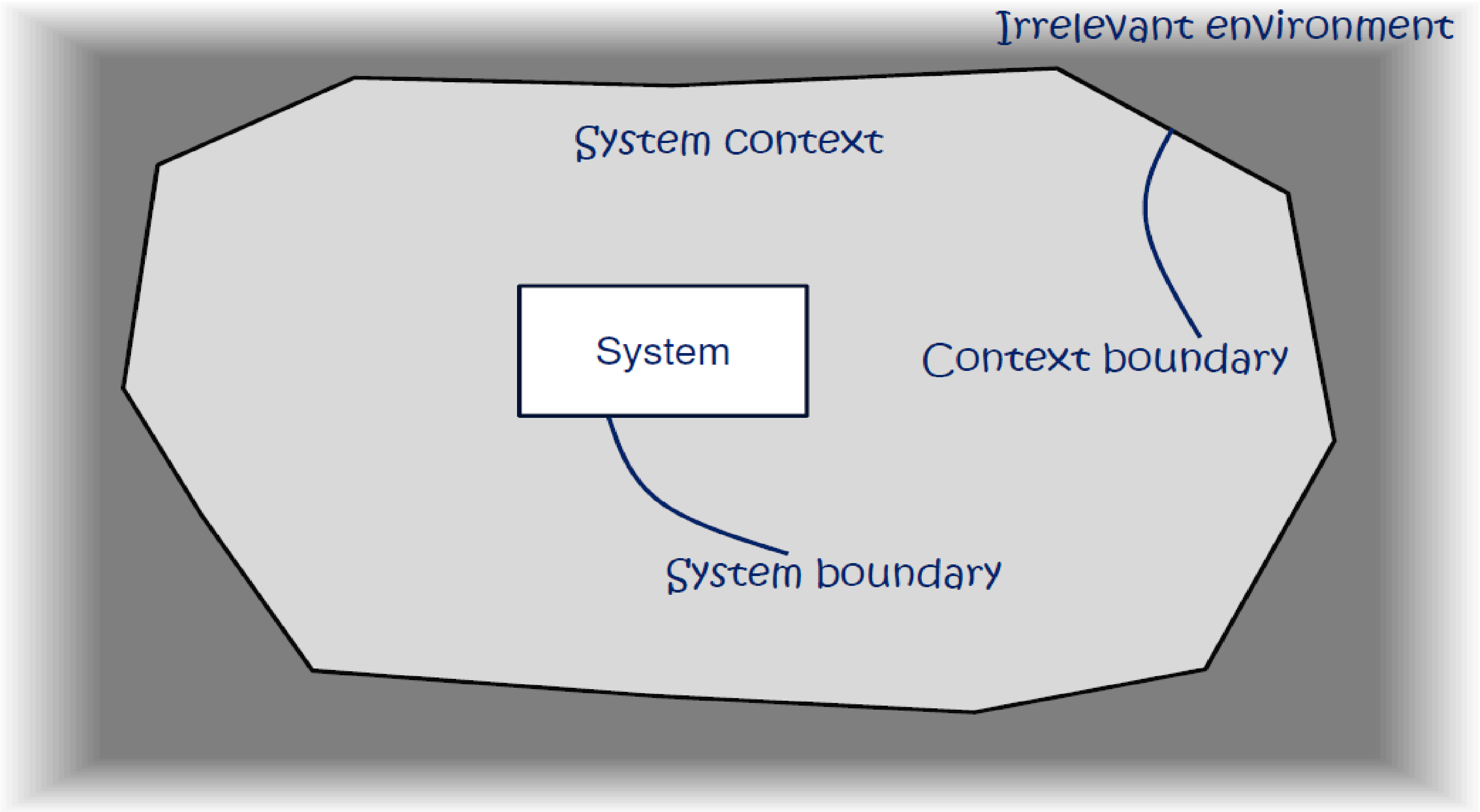
Summary

- A (software-intensive) system is always embedded in a particular context – the system context.
- A requirement is always defined for a particular context.
- The system context comprises system context objects.
- The system context is structured into three facets
  - Subject facet
  - Usage facet
  - IT system facet
- In addition to the context object, their properties and relationships have to be considered during requirements engineering.

System & Context Boundaries

Considering Boundaries

Considering Boudaries during system development



Questions for Self-Assessment

- Can a requirement be defined without knowing the context the system is embedded/operating in?
- What influence does the context has?
- What are system context objects?
- Describe typical examples of material and immaterial context objects.
- How can the system context be structured?
- What are the relations in this structure?
- Why should properties and relationships of context objects be considered?

Questions ??

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